

THOTA VENKATA VISHNU VARDHAN

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Data Science & Analytics graduate-to-be who builds data science solutions spanning machine learning, business intelligence, and natural language processing. Built and deployed end-to-end analytics applications using Python, Machine Learning, Power BI, and Streamlit, including customer churn prediction (AUC-ROC 0.8343), retail sales analytics, and large-scale NLP sentiment analysis across 568,454 Amazon reviews. IBM Data Science certified with strong foundations in exploratory data analysis, business intelligence, statistical modeling, and communicating data-driven insights to technical and non-technical stakeholders.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java
- **Data Analysis & Visualization:** SQL, Power BI, Tableau, Microsoft Excel (Pivot Tables, Power Query, VLOOKUP, XLOOKUP), Pandas, NumPy, Matplotlib, Seaborn, Plotly
- **Machine Learning:** Scikit-learn, Logistic Regression, Random Forest, Feature Engineering, Model Evaluation (AUC-ROC, Cross-Validation), SMOTE
- **Natural Language Processing:** NLTK, VADER, TextBlob, TF-IDF, WordCloud
- **Tools:** Git, GitHub, Streamlit, Jupyter Notebook
- **Core Analytics:** Exploratory Data Analysis (EDA), Data Cleaning, Data Wrangling, Data Visualization, Dashboard Development, Business Intelligence, Hypothesis Testing, A/B Testing, SQL Window Functions, KPI Reporting

PROJECTS

Marketing A/B Testing Analysis | Python, SQL (MySQL), Tableau, Git, GitHub

- Built a MySQL-to-Python pipeline for **588,101 marketing records**, resolving a silent import-truncation failure via staged **LOAD DATA INFILE** transformation and **SQL window functions** for hourly/daily trend aggregation.
- Applied a **two-proportion Z-test** to validate a **0.77-point conversion lift (p<0.001, 95% CI: 0.60%–0.94%)** over a PSA control group, translating results into an estimated **₹21.7 lakh revenue impact** via an interactive Tableau dashboard.

Customer Churn Prediction System | Python, Scikit-learn, Pandas, SMOTE, Streamlit, Git, GitHub

- Addressed a **73.5% / 26.5%** class imbalance by applying **SMOTE exclusively to the training dataset**, expanding training samples from **5,625 to 8,260**, then trained Logistic Regression and Random Forest models with Random Forest achieving an **AUC-ROC of 0.8343**.
- Identified that **month-to-month customers experienced 42.7% churn compared to only 2.9% for two-year contracts**, providing actionable retention insights through exploratory data analysis and feature importance evaluation.
- Deployed as a live web app on Streamlit Cloud — customer-churn-prediction-0001.streamlit.app

Product Review Sentiment Dashboard | Python, NLTK, VADER, TextBlob, TF-IDF, WordCloud, Plotly, Scikit-learn, Git, GitHub

- Built an end-to-end NLP sentiment analysis pipeline processing **568,454 Amazon Fine Food Reviews (1999–2012)** using **VADER, TextBlob**, and custom preprocessing, achieving **79.6% agreement** with ground-truth star ratings while extracting high-impact negative phrases using **TF-IDF**.
- Analyzed sentiment misclassification patterns and discovered that nearly **50% of one-star reviews** were classified as **POSITIVE** by VADER due to expectation-framing language, recommending a hybrid sentiment scoring approach for improved production reliability.
- Deployed as a live web app on Streamlit Cloud — product-sentiment-dashboard.streamlit.app

Rossmann Store Sales Forecasting Pipeline | Python, SQL, Pandas, NumPy, Prophet, XGBoost, Scikit-learn, Power BI, DAX, Excel

- Built a MySQL data pipeline processing **1M+ records** across **1,115 stores** using **joins, CTEs, and window functions** across seven analytical queries, identifying and resolving a **silent NULL-handling bug** that was fabricating data values.
- Benchmarked **Prophet** against **XGBoost** after tuning Prophet through **baseline, holiday-aware, and seasonality stages**, then selected **XGBoost** as the production model after **RandomizedSearchCV** tuning cut forecast error from **21.00% to 17.13% MAPE**, retaining Prophet for its **interpretable trend and seasonality decomposition**.

EDUCATION

B.Tech, Artificial Intelligence & Data Science — Study World College of Engineering | 2024–2027 | CGPA 7.72

Diploma, Computer Science — Govt. Polytechnic College | 2021–2024 | CGPA 8.10

CERTIFICATIONS

- [Google Cloud Data Analytics Certificate](#) — Google
- [IBM Data Science Professional Certificate](#) — Coursera
- Oracle Databases for Developers: Foundations — Oracle

ACHIEVEMENTS

- Earned 2★ SQL Badge on HackerRank by solving SQL programming and database query challenges.